Chemistry • Air pollution, AQI and climate

Environmental Chemistry

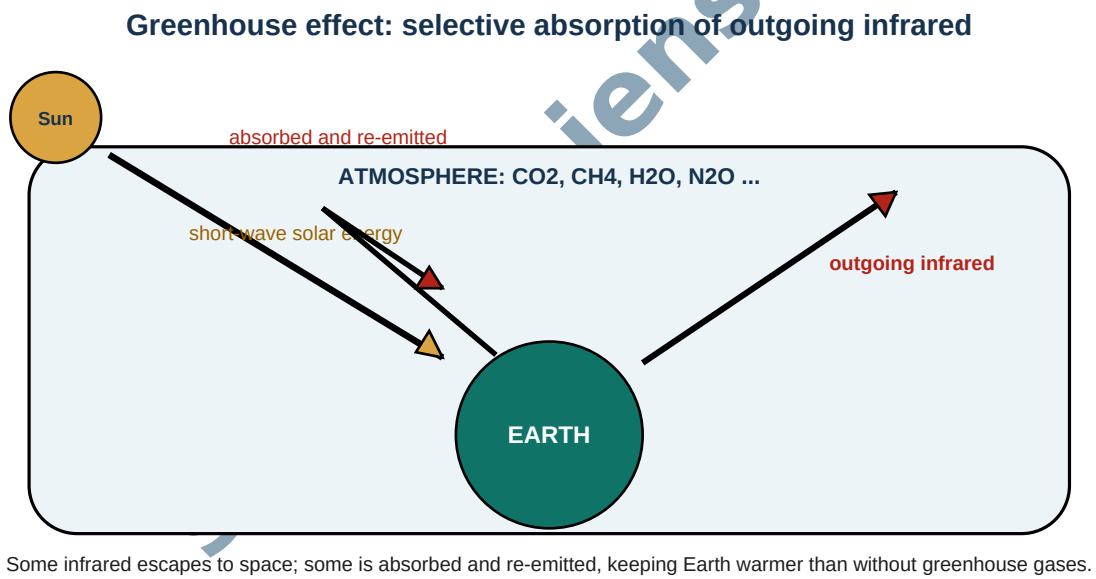
Official exercise • Complete solved answers • Pakistan climate investigation

Chapter 10 • Complete official exercise • 10 MCQs + 7 short + 5 constructed + 5 descriptive + 3 investigative

What this chapter is about

• **Air pollutants** include CO, SO_2, NO_x, particulate matter, VOCs and ground-level ozone; CO_2 and CH_4 drive human-enhanced global warming.
• **Acid rain** forms when sulphur and nitrogen oxides are converted to acids in the atmosphere.
• **The greenhouse effect** is natural and necessary, but additional emissions strengthen it and disturb climate.
• **Prevention** requires cleaner energy and transport, emission control, sustainable land/waste management, monitoring and protection of high-risk groups.

Figure 1 — Original schematic of the natural and enhanced greenhouse effect



Some infrared escapes to space; some is absorbed and re-emitted, keeping Earth warmer than without greenhouse gases.

Extra emissions strengthen this natural effect; N2 and O2 do not absorb infrared in the same way.

Exercise coverage: All five official sections are solved in book order. The Pakistan climate investigation uses a 2021–2026 evidence window and dated sources.

A. Multiple Choice Questions

1. Which gases are responsible for the greenhouse effect?

(a) SO_2, NO_x
(b) NO_x, CO
(c) CO_2, CH_4
(d) O_2, N_2

Answer: (c) CO_2, CH_4 — CO_2 and CH_4 absorb outgoing infrared radiation and are major anthropogenic greenhouse gases.

2. Indicate the source of sulphur responsible for oxides of sulphur in the atmosphere.

(a) decomposition of vegetation
(b) waste gases from digestion of animals
(c) photochemical smog
(d) combustion of fossil fuels

Answer: (d) combustion of fossil fuels — Sulphur in coal and petroleum oxidizes during combustion, releasing mainly SO_2.

3. Concentration of which atmospheric gas is decreased by photosynthesis in plants?

(a) oxygen
(b) nitrogen
(c) carbon dioxide
(d) water vapours

Answer: (c) carbon dioxide — Photosynthesis consumes CO_2 and water and releases oxygen while forming carbohydrate.

4. Which substance remains unaffected in the catalytic converter fixed in an automobile exhaust?

(a) CO_2
(b) CO
(c) NO
(d) NO_2

Answer: (a) CO_2 — CO and nitrogen oxides are catalytically converted, while CO_2 is an oxidation product and is not removed by the converter.

5. People of which age groups are most affected by air pollution?

(a) young adults
(b) middle-aged people
(c) children
(d) both children and aged people

Answer: (d) both children and aged people — Children have developing lungs and higher intake per body mass; older people more often have reduced reserve and chronic disease.

6. In which area is there a greater possibility of acid rain?

(a) around a village
(b) around big cities
(c) around an industrial area
(d) around water bodies

Answer: (c) around an industrial area — Industrial combustion and processing can emit large amounts of SO_2 and NO_x, the main acid-rain precursors.

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7. Why is smog not felt as much in summer?

(a) because fog is not present in summer
(b) because Earth's heat makes smoke rise
(c) because smoke and fog cannot mix in summer
(d) because less fossil fuel is burnt in summer

Answer: (b) because Earth's heat makes smoke rise — The intended answer is stronger convection: warm air and pollutants rise and disperse. This is a general tendency, not a guarantee of clean summer air.

8. Which catalysts are used in automobile catalytic converters?

(a) Ni
(b) Cu
(c) Pt, Pd and Rh
(d) CaO

Answer: (c) Pt, Pd and Rh — Platinum, palladium and rhodium catalyse oxidation of CO/hydrocarbons and reduction of nitrogen oxides.

9. Which components are essential for formation of photochemical smog?

(a) CO, NO_x, CO_2
(b) NO_x, volatile organic compounds, sunlight
(c) CO_2, NO_x, sunlight
(d) volatile organic compounds, NO_x, CO

Answer: (b) NO_x, volatile organic compounds, sunlight — Sunlight drives reactions of NO_x and VOCs that produce ozone and other photochemical oxidants.

10. Which gases contribute towards formation of acid rain?

(a) oxides of carbon
(b) oxides of sulphur
(c) oxides of nitrogen
(d) both oxides of nitrogen and sulphur

Answer: (d) both oxides of nitrogen and sulphur — SO_2/SO_3 and NO/NO_2 form sulphurous/sulphuric and nitrous/nitric acids in atmospheric moisture.

B. Questions for Short Answers

10.1. What is the main objective of environmental education?

Its objective is to develop knowledge, awareness, values and practical skills so people understand environmental systems, recognize human impacts and participate in sustainable solutions. It should lead to responsible behaviour and informed civic action, not just memorization of environmental terms.

10.2. How is particulate matter released into the atmosphere?

Particles enter directly from dust storms, soil erosion, sea spray, pollen and volcanic activity, and from vehicle exhaust, power plants, industry, brick kilns, construction, mining, agriculture and burning of waste or biomass. Some particles also form secondarily in air when gaseous SO_2, NO_x, ammonia and VOCs react.

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10.3. Which gas is more poisonous, CO_2 or CO?

Carbon monoxide (CO) is much more acutely poisonous. It binds strongly to haemoglobin as carboxyhaemoglobin and reduces oxygen transport, causing headache, unconsciousness or death. CO_2 is not similarly poisonous at normal atmospheric levels, though very high concentrations can displace oxygen and both gases have environmental effects.

10.4. How does acid rain affect forests?

Acid rain damages leaves and needles, leaches Ca, Mg and K nutrients from soil, and mobilizes aluminium that can injure roots. It reduces nutrient uptake and growth and makes trees more vulnerable to drought, frost, insects and disease. Acidified runoff can also harm aquatic food webs on which forest ecosystems depend.

10.5. In what way does sulphur present in fossil fuels become dangerous?

During combustion, sulphur forms sulphur dioxide, which irritates airways and reacts further in the atmosphere to produce acid-rain acids and sulphate particles.

$$S + O_2 \rightarrow SO_2$$

$$2SO_2 + O_2 \rightarrow 2SO_3;\ SO_3 + H_2O \rightarrow H_2SO_4$$

$$SO_2 + H_2O \rightleftharpoons H_2SO_3$$

The resulting pollution harms lungs, vegetation, soils, lakes, buildings and metal structures. Low-sulphur fuels, scrubbers and cleaner energy reduce these emissions.

10.6. Name any three major sources responsible for the greenhouse effect.

Three major human-related sources are: **(1) fossil-fuel combustion** for electricity, transport and industry, releasing CO_2; **(2) agriculture**, especially livestock and rice production releasing CH_4 and fertilizer/soil processes releasing N_2O; and **(3) deforestation and land-use change**, which release stored carbon and reduce CO_2 absorption. Cement production and landfill waste are additional sources.

10.7. How is wind energy useful for us?

Wind turns turbine blades; the rotor and generator convert kinetic energy into electricity. It is renewable and produces no direct smoke, SO_2, NO_x or CO_2 during operation, making it useful for grid supply and remote areas. Its limitations include variable wind, the need for backup/storage, land and visual impacts, noise and possible effects on birds and bats.

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C. Constructed Response Questions

10.1. How is excessive use of insecticides and pesticides harmful for birds?

Birds can be poisoned directly or by eating contaminated insects, seeds, fish or small animals. Persistent chemicals accumulate through food chains and may cause eggshell thinning, reproductive failure, behavioural or immune damage and death. Pesticides also reduce insect prey and biodiversity, while spray drift and runoff expand exposure beyond farms.

10.2. CO_2 is only about 0.04% of air. How can it become harmful for the ecosystem?

Even a small atmospheric fraction can have a large radiative effect. CO_2 absorbs outgoing infrared radiation at specific wavelengths; N_2 and O_2, which make up most air, do not absorb it in the same way.

Human emissions add to natural sources faster than forests, soils and oceans can remove the excess. CO_2 therefore accumulates, and warming is amplified by feedbacks involving water vapour, ice loss and clouds.

The resulting climate shifts alter rainfall, heat extremes, glaciers, sea level and food production. Increased ocean CO_2 also causes **ocean acidification**, affecting shell-forming organisms. Thus harm depends on radiative potency, persistence and cumulative increase—not simply on percentage abundance.

10.3. Why do only some atmospheric pollutant gases cause the greenhouse effect while others do not?

A greenhouse gas must absorb infrared radiation. Molecules such as CO_2, CH_4, H_2O and N_2O have vibrations that change their charge distribution and interact strongly with infrared wavelengths, after which they re-emit energy in all directions.

Homonuclear diatomic molecules such as N_2 and O_2 lack the necessary infrared-active vibration and are effectively transparent to this radiation.

A gas's actual warming contribution also depends on its absorption bands, atmospheric lifetime, concentration and whether other gases already absorb at those wavelengths.

10.4. How can CO emissions from automobile fuel combustion be reduced?

• **Maintain and tune the engine:** Clean injectors/plugs, working sensors and the correct air–fuel ratio promote complete combustion.

• **Ensure sufficient oxygen:** Proper air intake and modern electronic fuel injection reduce fuel-rich operation.

• **Use a functioning catalytic converter:** $2CO + O_2 \rightarrow 2CO_2$.

• **Use cleaner fuels and technology:** Low-carbon fuels, efficient engines, hybrids or electric vehicles reduce exhaust CO.

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• **Change driving practice:** Avoid unnecessary idling and use inspection, public transport or efficient route planning.

CO forms mainly through incomplete combustion, so both combustion control and exhaust after-treatment are important.

10.5. Mention three different ways in which solar energy can be useful for us.

1. Solar electricity: Photovoltaic cells power homes, schools, streetlights, communication equipment and water pumps.

2. Solar heating: Solar collectors heat water or indoor spaces, reducing gas and electricity demand.

3. Solar thermal processing: Solar cookers, dryers and distillers cook food, preserve crops and obtain clean water.

These uses reduce fossil-fuel consumption and local emissions, although initial cost, storage and weather variability must be managed.

D. Descriptive Questions

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10.1. Describe the harmful effects of the major pollutants present in the air.

Pollutant	Major sources	Harmful effects
Carbon monoxide, CO	Incomplete petrol, gas, wood and coal combustion	Reduces blood oxygen transport; dizziness, heart stress, unconsciousness or death.
Sulphur dioxide, SO_2	Sulphur-containing coal/oil, smelting and industry	Airway irritation, asthma aggravation, acid rain and sulphate particles.
Nitrogen oxides, NO_x	High-temperature vehicle and power-plant combustion	Respiratory irritation, acid rain, nitrate particles and photochemical smog.
Particulate matter, PM_{10}/$PM_{2.5}$	Dust, smoke, exhaust, kilns, construction and secondary reactions	Penetrates airways; $PM_{2.5}$ reaches deep lung/blood and increases heart and lung disease.
Ground-level ozone, O_3	Sunlight-driven NO_x + VOC reactions	Chest pain, reduced lung function, asthma attacks and crop damage.
VOCs	Fuel evaporation, solvents, industry and incomplete combustion	Smog formation; some are toxic or carcinogenic, e.g. benzene.
CO_2 and CH_4	Fossil fuels, agriculture, waste and land-use change	Enhanced greenhouse warming and climate disruption; CH_4 also supports ozone formation.

Effects often occur together, and long-term exposure to low concentrations can be as important as a single severe episode.

10.2. Explain the greenhouse effect. How is global warming dangerous for us?

Solar short-wave energy reaches Earth, which absorbs much of it and emits infrared radiation. Greenhouse gases absorb and re-emit part of this outgoing energy. The natural effect keeps Earth warm enough for life; additional CO_2, CH_4, N_2O and other gases strengthen it and produce an energy imbalance.

• **Health:** More heatstroke, cardiovascular stress, vector-borne disease and smoke/dust exposure.

• **Water and food:** Changed monsoon timing, droughts, floods, lower crop yields and livestock stress.

• **Ice and coasts:** Glacier retreat alters river flows; thermal expansion and melting land ice raise sea level.

• **Extreme events:** Greater heat extremes and heavier rainfall episodes increase disaster risk.

• **Ecosystems and economy:** Species loss, coral damage, infrastructure destruction and livelihood insecurity.

Mitigation reduces further emissions; adaptation—early warning, resilient crops, water management and safer infrastructure—reduces harm from changes already occurring.

10.3. What is the Air Quality Index? What information does it convey?

The **Air Quality Index (AQI)** converts measured concentrations of selected pollutants into a numerical category for public communication. Agencies calculate sub-indices for pollutants such as $PM_{2.5}$, PM_{10}, O_3, CO, SO_2 and NO_2; the highest relevant sub-index usually determines the reported AQI.

AQI information	What it tells the public
Number / category and colour	Relative level of air pollution, such as good, moderate, unhealthy or hazardous
Dominant pollutant	Which measured pollutant is driving the warning
Health guidance	Risk to sensitive and general groups and whether outdoor activity should be reduced
Time and location	When and where the measurement or forecast applies

AQI scales and breakpoints vary by country and agency. A single AQI does not identify every chemical or prove safety for every person, so the dominant pollutant, time and local advisory should also be read.

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10.4. Who are high-risk groups and why is pollution more dangerous for them?

Group	Reason for greater risk
Children	Lungs and immune systems are developing; they breathe more air per kilogram and spend time active outdoors.
Older people	Reduced heart/lung reserve and more chronic illness increase complications.
Asthma, COPD and heart-disease patients	Particles, ozone and gases can trigger attacks, arrhythmias or hospitalization.
Pregnant people	Pollution exposure can affect both maternal health and fetal development.
Outdoor workers and athletes	Longer exposure and deeper breathing increase inhaled dose.
Low-income or highly exposed communities	Greater proximity to traffic, industry, indoor smoke and limited healthcare increases risk.

High-risk groups should follow local AQI alerts, reduce exertion during severe episodes, use cleaner indoor-air practices and seek medical advice for symptoms.

10.5. Describe three strategies to address environmental issues.

1. Prevent emissions and resource damage: Shift to renewable energy, efficient industry, clean public transport, low-sulphur fuels, methane control, sustainable agriculture and reduced waste.

2. Monitor and enforce: Use continuous AQI/water monitoring, emission standards, vehicle inspection, industrial scrubbers and catalytic controls, transparent data and effective penalties.

3. Restore ecosystems and protect people: Plant suitable native trees, restore wetlands and drainage, manage waste, provide early warnings and health guidance, and involve schools and communities in adaptation.

The strongest programmes combine technology, law, finance and public participation rather than relying on tree planting or cleanup alone.

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E. Investigative Questions

10.1. Major Pakistani cities experience very high AQI in winter. Point out major causes.

• **Meteorology:** Temperature inversions trap a shallow layer of cold air near the ground; weak winds, low mixing height, fog and little rain prevent dispersion.

• **Transport:** Old or poorly maintained vehicles, diesel exhaust, congestion and evaporative fuel emissions add PM, NO_x, CO and VOCs.

• **Industry and energy:** Brick kilns, furnaces, generators and coal/biomass use emit SO_2, NO_x and particles.

• **Open burning and dust:** Crop-residue and municipal-waste burning, construction and unpaved-road dust increase particulate loading.

• **Secondary pollution:** NO_x, SO_2, VOCs and ammonia form additional fine particles; regional transport can carry pollution between cities and across borders.

Winter smog is therefore caused by **emissions plus trapping weather**; fog alone does not create the toxic pollutants.

10.2. Why does AQI not rise in Pakistan on hot summer days?

The premise is only partly true. Hot ground often produces stronger convection and a deeper mixing layer, while monsoon winds and rain can disperse or wash out particles. There is also less winter inversion and, in some areas, less seasonal biomass/residue burning. These factors can lower particulate AQI.

However, summer AQI **can** become unhealthy: strong sunlight forms ground-level ozone from NO_x and VOCs, dust storms raise PM, heat waves increase energy-generator use, and dry pre-monsoon weather can trap smoke locally.

Thus “summer is always low AQI” is incorrect; actual AQI depends on pollutant emissions, weather, location and the pollutant being measured.

10.3. How has climate change affected Pakistan during the last five years?

During the **2021–2026 evidence window**, Pakistan has experienced interacting heat, rainfall and flood hazards rather than a single uniform trend:

1. 2022 catastrophic floods: Exceptional monsoon rainfall and runoff affected about 33 million people, caused over 1,700 deaths and damaged homes, crops and infrastructure. UNDP reports damages and economic losses exceeding US$30 billion.

2. 2024 extreme heat: Repeated heatwaves brought forecasts/readings near 50°C in parts of Pakistan, disrupted schooling and increased heatstroke risk. Pakistan’s April 2024 was also reported as its wettest since 1961, showing the variability between extremes.

3. 2025 monsoon flooding: UN reporting described over six million people affected and nearly 1,000 deaths by September, with major displacement and crop/infrastructure losses in Punjab, Khyber Pakhtunkhwa and Gilgit-Baltistan.

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Recurring consequences include food and water insecurity, glacier-related hazards, disease outbreaks, heat stress, urban flooding and pressure on disaster-response and public finances. Climate change increases the likelihood or severity of many such extremes, while exposure, drainage, land use and poverty influence the final damage.

Dated evidence: UNDP Pakistan, "Building Resilience: UNDP's Response to Pakistan's Recurring Floods" (28 Oct 2025, undp.org); UN News, "The needs are huge: Pakistan reels from floods" (16 Sep 2025, news.un.org); VOA reports citing the Pakistan Meteorological Department on extreme heat (22–23 May 2024, voanews.com).

Points to Remember

CO is acutely toxic through haemoglobin binding; **CO_2 and CH_4** are major greenhouse gases.

SO_2 and NO_x cause respiratory harm, particles and acid rain; NO_x + VOCs + sunlight produce photochemical smog.

$PM_{2.5}$ is especially dangerous because it penetrates deeply into lungs and can enter circulation.

The greenhouse effect is natural; additional emissions enhance it. Gas behaviour depends on infrared absorption, not simply concentration.

Catalytic converters reduce CO, hydrocarbons and NO_x but do not remove CO_2.

AQI summarizes selected pollutants and health guidance; it is not a complete chemical analysis of air.

Winter smog results from emissions plus inversions, weak mixing and low washout; summer can still have ozone and dust pollution.

Solutions combine clean energy, enforcement, ecosystem protection, public information and protection of vulnerable people.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	c	2	d	3	c	4	a	5	d
6	c	7	b	8	c	9	b	10	d

How to attempt this chapter in the paper

- Separate local health pollutants (CO, PM, SO_2, NO_x, ozone) from long-lived climate gases (CO_2, CH_4, N_2O).
- For "cause" questions, include both emissions and the atmospheric process—oxidation, photochemistry, inversion or infrared absorption.
- For AQI answers, mention dominant pollutant, category, location/time and health advice rather than quoting a number alone.
- For current investigations, date the examples and avoid claiming that one flood or heatwave has a single cause.

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